

Roll and Pitch Observability in a Local IMU/GNSS EKF

sensor_fusion documentation

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1 Scope

This note formulates roll and pitch observability for a generic local error-state EKF using the project's Reduced NED convention. The same local analysis applies to the Full ECEF filter after projecting ECEF velocity and attitude into the local vehicle frame. The purpose is to show why vehicle attitude and mount attitude are strongly coupled, and why non-holonomic constraints (NHC), velocity observations, and excitation are required to separate them.

The analysis is intentionally local and first-order. It does not replace the generated Jacobians used by the filters. It provides a compact model for reasoning about which residuals can distinguish vehicle attitude from physical mounting error.

2 Generic Local EKF Context

The local EKF convention is the Reduced convention:

$$x_n = C_{nv}x_v, \quad x_v = C_{nv}^T x_n, \quad (1)$$

where n is local NED and v is the vehicle frame. The vehicle frame is forward-right-down. The raw IMU body frame is b , and the physical mount is

$$x_b = C_{bv}x_v, \quad x_v = C_{bv}^T x_b. \quad (2)$$

The nominal state relevant to this note is

$$x = \begin{bmatrix} q_{nv} & v_n^T & p_n^T & b_g^T & b_a^T & q_{bv} \end{bmatrix}^T. \quad (3)$$

The corresponding error-state components are vehicle attitude error, navigation velocity error, position error, IMU bias error, and mount error:

$$\delta x = \begin{bmatrix} \delta\theta_v & \delta v_n & \delta p_n & \delta b_g & \delta b_a & \delta\psi_{bv} \end{bmatrix}^T. \quad (4)$$

For local observability arguments we project velocity and acceleration errors into the vehicle frame:

$$\delta v_v \approx C_{nv}^T \delta v_n. \quad (5)$$

Full uses $v_v = C_{ev}^T v_e$, but once this projection is made the local roll/pitch ambiguity has the same structure. Therefore the rest of the document uses the Reduced NED notation.

2.1 Measurements

The key measurements and pseudo-measurements are:

$$\text{GNSS velocity: } z_{v_n} = v_n + \eta_{v_n}, \quad (6)$$

$$\text{NHC lateral: } z_y = 0 = e_y^T v_v + \eta_y, \quad (7)$$

$$\text{NHC vertical: } z_z = 0 = e_z^T v_v + \eta_z, \quad (8)$$

$$\text{optional vehicle speed: } z_x = v_x + \eta_x. \quad (9)$$

GNSS velocity anchors the velocity states. NHC says that a ground vehicle should have near-zero lateral and vertical velocity in the vehicle frame. These measurements constrain attitude and mount only through their effect on velocity propagation and the projection from navigation velocity into the vehicle frame.

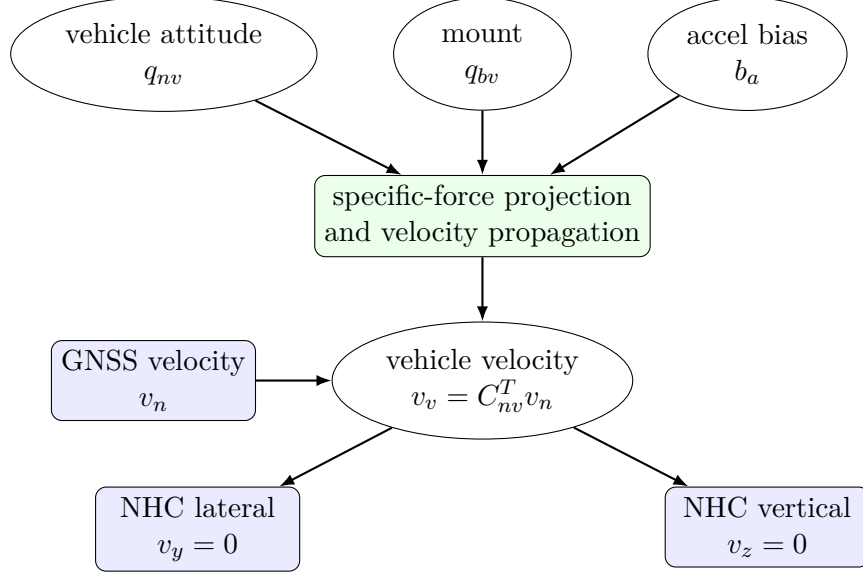


Figure 1: Generic local EKF context. Attitude, mount, and accelerometer bias affect velocity propagation. GNSS velocity and NHC then determine whether those effects can be separated.

3 Roll Channel

3.1 Roll-Only State

Use the roll-only local error state

$$x_r = [\delta\phi_v \quad \delta\rho_b \quad \delta b_{ay} \quad \delta b_{az} \quad \delta v_y \quad \delta v_z]^T. \quad (10)$$

Here $\delta\phi_v$ is vehicle roll error, $\delta\rho_b$ is vehicle-to-body mount roll error, δb_{ay} and δb_{az} are lateral and vertical accelerometer correction errors after local projection, and $\delta v_y, \delta v_z$ are vehicle-frame lateral and vertical velocity errors.

3.2 Roll Dynamics

For small roll errors, and ignoring second-order yaw/pitch products, the lateral and vertical velocity error dynamics are approximately

$$\delta\dot{v}_y \approx g\delta\phi_v - g\delta\rho_b - \delta b_{ay} + w_y, \quad (11)$$

$$\delta\dot{v}_z \approx -a_y\delta\rho_b - \delta b_{az} + w_z. \quad (12)$$

The sign convention can flip with implementation details, but the structure is the invariant point: the lateral channel observes the difference between vehicle roll and mount roll, while the vertical channel contains a mount-only term proportional to lateral acceleration a_y .

3.3 Roll NHC Rows

Over one short interval Δt , Equations (11) and (12) give

$$r_y \approx -\delta v_y - \Delta t(g\delta\phi_v - g\delta\rho_b - \delta b_{ay}) + \eta_y, \quad (13)$$

$$r_z \approx -\delta v_z - \Delta t(-a_y\delta\rho_b - \delta b_{az}) + \eta_z. \quad (14)$$

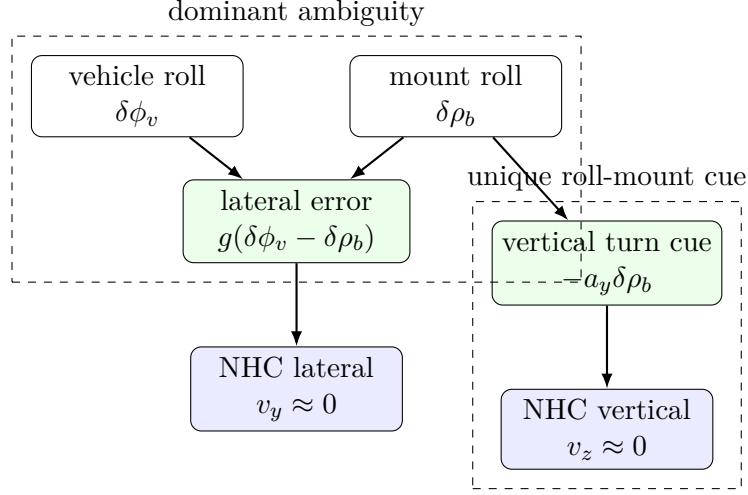


Figure 2: Roll ambiguity. The strongest roll signal is lateral gravity projection, but it sees $\delta\phi_v - \delta\rho_b$. The unique mount-roll cue enters through vertical NHC during turns.

The corresponding Jacobian rows are

$$H_y \approx \begin{bmatrix} -g\Delta t & +g\Delta t & +\Delta t & 0 & -1 & 0 \end{bmatrix}, \quad (15)$$

$$H_z \approx \begin{bmatrix} 0 & +a_y\Delta t & 0 & +\Delta t & 0 & -1 \end{bmatrix}. \quad (16)$$

Thus lateral NHC is strong but ambiguous. Vertical NHC is weaker but separates mount roll from vehicle roll when a_y is nonzero.

4 Pitch Channel

4.1 Pitch-Only State

Use the pitch-only local error state

$$x_p = \begin{bmatrix} \delta\theta_v & \delta\pi_b & \delta b_{ax} & \delta b_{az} & \delta v_x & \delta v_z \end{bmatrix}^T. \quad (17)$$

Here $\delta\theta_v$ is vehicle pitch error, $\delta\pi_b$ is vehicle-to-body mount pitch error, δb_{ax} and δb_{az} are longitudinal and vertical accelerometer correction errors after local projection, and $\delta v_x, \delta v_z$ are vehicle-frame longitudinal and vertical velocity errors.

4.2 Pitch Dynamics

For small pitch errors, the longitudinal and vertical velocity error dynamics are approximately

$$\delta\dot{v}_x \approx -g\delta\theta_v + g\delta\pi_b - \delta b_{ax} + w_x, \quad (18)$$

$$\delta\dot{v}_z \approx +a_x\delta\pi_b - \delta b_{az} + w_z. \quad (19)$$

The first row is the pitch analogue of the roll lateral row: it mainly observes the difference between vehicle pitch and mount pitch. The second row is the unique mount-pitch cue. It is driven by longitudinal acceleration a_x , so acceleration and braking maneuvers provide the excitation that separates mount pitch from vehicle pitch.

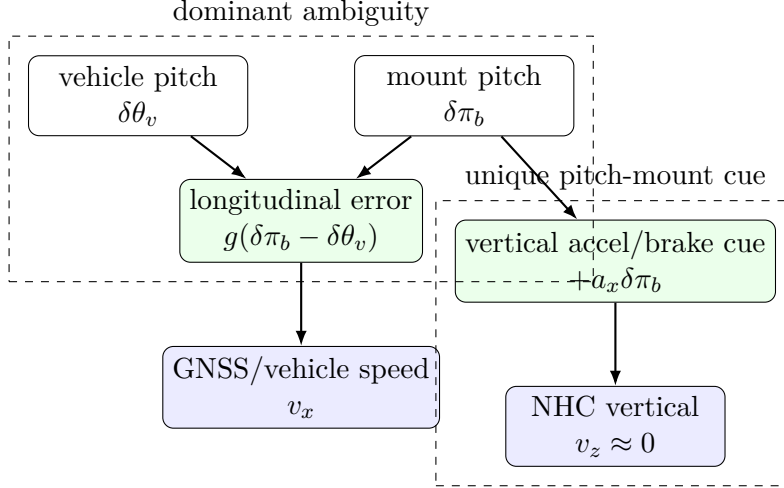


Figure 3: Pitch ambiguity. Vehicle pitch and mount pitch trade through longitudinal gravity projection. Mount pitch becomes uniquely observable when longitudinal acceleration or braking excites the vertical channel.

4.3 Pitch Measurement Rows

There is no ordinary NHC constraint on longitudinal velocity, because v_x is the intended vehicle motion. Longitudinal velocity must therefore be anchored by GNSS velocity, an optional vehicle-speed measurement, or the integrated trajectory. A short-interval longitudinal velocity residual is

$$r_x \approx -\delta v_x - \Delta t(-g\delta\theta_v + g\delta\pi_b - \delta b_{ax}) + \eta_x. \quad (20)$$

The vertical NHC row is

$$r_z \approx -\delta v_z - \Delta t(a_x\delta\pi_b - \delta b_{az}) + \eta_z. \quad (21)$$

With state order $x_p = [\delta\theta_v, \delta\pi_b, \delta b_{ax}, \delta b_{az}, \delta v_x, \delta v_z]^T$, the rows are

$$H_x \approx \begin{bmatrix} +g\Delta t & -g\Delta t & +\Delta t & 0 & -1 & 0 \end{bmatrix}, \quad (22)$$

$$H_z \approx \begin{bmatrix} 0 & -a_x\Delta t & 0 & +\Delta t & 0 & -1 \end{bmatrix}. \quad (23)$$

The pitch analogue of roll's lateral NHC row is the longitudinal velocity row. It is not a zero-velocity pseudo-measurement; it comes from GNSS velocity, vehicle speed, or velocity consistency. The unique pitch-mount information is again the vertical NHC row, now scaled by a_x .

5 Information Strength

For roll, the approximate ratio of unique mount-roll information to coupled roll information is

$$\frac{I_{\rho,\text{unique}}}{I_{\rho,\text{coupled}}} \approx \frac{a_y^2/R_z}{g^2/R_y} = \left(\frac{a_y}{g}\right)^2 \frac{R_y}{R_z}. \quad (24)$$

For pitch, the analogous ratio is

$$\frac{I_{\pi,\text{unique}}}{I_{\pi,\text{coupled}}} \approx \frac{a_x^2/R_z}{g^2/R_x} = \left(\frac{a_x}{g}\right)^2 \frac{R_x}{R_z}, \quad (25)$$

where R_x is the effective longitudinal velocity observation variance from GNSS, vehicle speed, or the relevant batch update. If the longitudinal velocity anchor is weak, pitch can still trade with mount pitch even when R_z is tight.

Using representative NHC values $R_y = 0.5$, $R_z = 0.05$, the roll ratio is

$$10 \left(\frac{a_y}{9.80665} \right)^2. \quad (26)$$

The same scale applies to pitch if $R_x \approx R_y$. In practice R_x is often different because forward velocity is measured rather than constrained to zero.

Excitation [m/s ²]	Maneuver	Ratio with coupled/vertical $R = 10$	Interpretation
0.5	mild	0.026	very weak unique cue
1.0	moderate	0.104	weak unique cue
2.0	strong	0.416	useful but still coupled
3.0	very strong	0.936	comparable to coupled cue

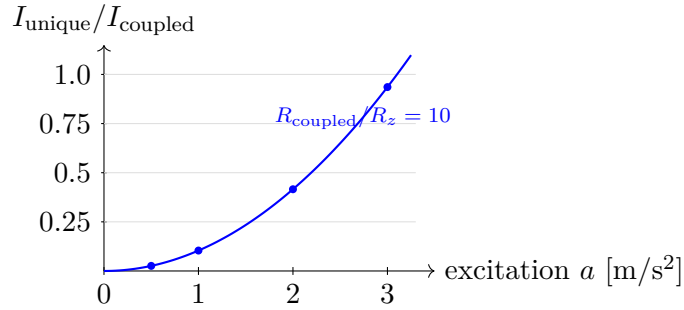


Figure 4: Unique mount-angle information is weak under mild excitation. Roll uses lateral acceleration a_y ; pitch uses longitudinal acceleration a_x .

6 Bias and Covariance Effects

The ratios above are optimistic because they ignore bias. A one-degree attitude or mount error projects gravity into a horizontal axis by

$$g \sin(1^\circ) \approx 0.171 \text{ m/s}^2. \quad (27)$$

The unique vertical cue from one degree of mount error during a 1 m/s^2 maneuver is only

$$1.0 \sin(1^\circ) \approx 0.017 \text{ m/s}^2. \quad (28)$$

That term is an order of magnitude smaller. It can be absorbed by vertical accelerometer bias or vertical velocity unless covariance and process noise prevent those nuisance states from explaining persistent maneuver-correlated residuals.

For a linearized update with information matrix

$$\Lambda = H^T R^{-1} H, \quad (29)$$

the information in a mount angle α after marginalizing nuisance states ξ is the Schur complement

$$\Lambda_{\alpha|\xi} = \Lambda_{\alpha\alpha} - \Lambda_{\alpha\xi} \Lambda_{\xi\xi}^{-1} \Lambda_{\xi\alpha}. \quad (30)$$

For roll,

$$\alpha = \delta\rho_b, \quad \xi = \begin{bmatrix} \delta\phi_v & \delta b_{ay} & \delta b_{az} & \delta v_y & \delta v_z \end{bmatrix}^T. \quad (31)$$

For pitch,

$$\alpha = \delta\pi_b, \quad \xi = \begin{bmatrix} \delta\theta_v & \delta b_{ax} & \delta b_{az} & \delta v_x & \delta v_z \end{bmatrix}^T. \quad (32)$$

This is the clean quantitative measure of whether the data window uniquely informs the mount angle after attitude, velocity, and bias are allowed to absorb the same residual.

7 Implications

1. Vehicle roll and mount roll are nearly interchangeable in the lateral channel. They are separated mainly by vertical NHC during lateral acceleration.
2. Vehicle pitch and mount pitch are nearly interchangeable in the longitudinal channel. They are separated mainly by vertical NHC during acceleration and braking.
3. GNSS velocity is necessary because it prevents v_y , v_z , and v_x from becoming hidden residual reservoirs, but it does not by itself make mount and vehicle attitude unique.
4. Bias covariance matters because the unique vertical cue is small. If vertical accelerometer bias is too flexible, it can absorb the same residual that should correct mount.
5. Mount covariance must remain flexible until enough uniquely informative excitation has accumulated. Premature covariance collapse makes the filter retain an incorrect roll/pitch split.

8 Recommended Replay Diagnostic

For each replay window, compute the linearized rows for NHC and velocity observations in the local Reduced basis and accumulate

$$\Lambda_W = \sum_{k \in W} H_k^T R_k^{-1} H_k. \quad (33)$$

Then report the raw and marginalized mount information:

$$I_{\alpha, \text{raw}} = \Lambda_{\alpha\alpha}, \quad (34)$$

$$I_{\alpha|\beta} = \Lambda_{\alpha\alpha} - \Lambda_{\alpha\beta} \Lambda_{\beta\beta}^{-1} \Lambda_{\beta\alpha}, \quad (35)$$

$$I_{\alpha|\xi} = \Lambda_{\alpha\alpha} - \Lambda_{\alpha\xi} \Lambda_{\xi\xi}^{-1} \Lambda_{\xi\alpha}. \quad (36)$$

For roll, $\alpha = \delta\rho_b$ and $\beta = \delta\phi_v$. For pitch, $\alpha = \delta\pi_b$ and $\beta = \delta\theta_v$. The raw value shows apparent mount information. The second value removes the vehicle-attitude ambiguity. The third value removes attitude, velocity, and accelerometer-bias nuisance states and is the most useful scalar observability metric.

9 Implemented Synthetic Experiment

The repository includes a repeatable diagnostic that exercises this theory in the Reduced NED convention:

```
cargo run --release -p sim --bin diag_mount_observability
```

The diagnostic generates truth-noise synthetic scenarios with the same injected initial mount error, computes the Schur-complement information proxies above, and then runs the Reduced EKF with two covariance splits:

- `qerr_mnt_loose`: tighter vehicle-attitude covariance and looser mount covariance,
- `qerr_att_loose`: looser vehicle-attitude covariance and tighter mount covariance.

If the data do not uniquely separate vehicle attitude from mount attitude, the final mount estimate depends strongly on this covariance split. That covariance sensitivity is itself evidence of weak observability. If a maneuver provides strong unique excitation, the Schur-complement information should increase for the corresponding axis: lateral acceleration for roll, and longitudinal acceleration/braking for pitch.